DETAILED PROBLEM STATMENT

PROBLEM STATEMENT

HONEY CHAIN: A BLOCKCHAIN-BASED SYSTEM FOR HONEY TRACEABILITY AND SMART BEEKEEPING MANAGEMENT

1. Introduction

Honey is a natural product widely consumed because of its nutritional value, distinctive taste, and applications in food and healthcare products. The production of honey involves several activities, including maintaining beehives, monitoring bee colonies, harvesting honey, processing, quality testing, packaging, storage, transportation, and distribution. Each stage contributes to the quality, authenticity, and safety of the final product. Therefore, maintaining accurate information throughout the honey supply chain is essential for producers, distributors, retailers, and consumers.

In traditional honey production systems, many beekeepers and honey producers maintain records manually or use separate digital applications to manage their activities. Information about hive locations, harvesting dates, honey quantities, quality inspection results, and distribution details may be recorded in different formats. These records are not always connected to one another, making it difficult to trace a particular honey product from its original source to the final consumer. When information is incomplete, outdated, or inconsistent, verifying the history of a honey batch becomes challenging.

The absence of a unified traceability system can also create difficulties when consumers want to know where their honey originated, when it was harvested, how it was processed, and whether its quality was tested. Similarly, producers and distributors may struggle to identify the source of a particular batch when a quality concern or customer complaint occurs. These challenges demonstrate the need for a digital platform that can maintain reliable product records and provide transparent access to relevant information.

The proposed **Honey Chain** project aims to address these problems by integrating blockchain-based traceability with smart beekeeping management. The system will create digitally linked records for beehives, honey harvesting, batch identification, quality testing, processing, packaging, and distribution. Blockchain technology will help preserve a tamper-evident history of important supply-chain events, while a digital management interface will allow beekeepers to organize their activities more efficiently.

The system will also provide QR code-based product verification, enabling consumers to access available honey-origin and supply-chain information using a smartphone. By combining digital record management, controlled information sharing, and blockchain-based traceability, Honey Chain aims to improve transparency, accountability, and trust throughout the honey supply chain.

2. Problems in the Existing Honey Supply Chain

The traditional honey supply chain faces several challenges related to information management, product traceability, and beekeeping operations. These challenges can affect operational efficiency and reduce the ability of stakeholders to verify product information.

2.1 Lack of End-to-End Honey Traceability

One of the major problems in the existing system is the difficulty of tracking honey throughout its complete journey. Honey passes through multiple stages, beginning with collection by beekeepers and continuing through processing, packaging, transportation, distribution, and retail sales. Information generated at each stage may be stored separately.

When there is no common traceability platform, connecting these records becomes difficult. For example, a retailer may know the batch number of a honey product but may not have immediate access to the corresponding harvesting information or apiary details. Similarly, a consumer may be unable to determine the source of the honey or identify the stages through which the product passed.

This lack of traceability can delay investigations when quality problems arise. It can also make it more difficult for producers to identify affected batches and provide accurate information to customers. Therefore, the proposed system must maintain a consistent relationship between the source hive, harvest record, honey batch, processing history, and distribution events.

2.2 Difficulty in Verifying Honey Authenticity

Honey authenticity is an important concern for consumers and businesses. Product labels may provide information about the producer, source, or quality of honey, but consumers may not have an easy way to verify these claims independently. Incomplete documentation and disconnected records can make verification more difficult.

A digital traceability system can help address this problem by connecting product information to identifiable records. In Honey Chain, every registered honey batch will receive a unique batch identification number. The batch record will be associated with relevant information, such as its source apiary, harvesting date, producer, processing details, and available quality test results.

A QR code linked to the batch record will allow consumers to view this information. However, digital traceability does not automatically prove the chemical purity of honey. Authenticity and quality claims must be supported by appropriate laboratory

testing, inspection procedures, and trustworthy data entry. The system will therefore distinguish between information that has been entered, information that has been verified, and information that remains pending verification.

2.3 Inefficient Beekeeping Record Management

Beekeeping requires regular monitoring and documentation of hive-related activities. Beekeepers may need to record hive identification numbers, locations, colony observations, harvesting schedules, honey quantities, and maintenance activities. When these details are maintained in notebooks or separate spreadsheets, managing a large number of hives can become time-consuming.

Manual records may be misplaced, duplicated, or updated inconsistently. Comparing production information across different hives or harvesting periods can also require additional effort. These limitations may affect planning and make it difficult to maintain an organized history of beekeeping activities.

Honey Chain proposes a digital interface through which beekeepers can register apiaries and individual hives, record harvesting information, and review previous production records. Each harvest entry will be linked to the corresponding hive so that the origin of a honey batch can be identified more easily. The system will help reduce dependence on scattered records and provide a structured method for maintaining beekeeping information.

2.4 Lack of Transparency Among Supply-Chain Participants

The honey supply chain involves several stakeholders, including beekeepers, producers, quality inspectors, processors, distributors, retailers, and consumers. Each stakeholder is responsible for particular activities and generates information during the movement of honey from one stage to another.

When these participants use separate systems, sharing information can become difficult. Some stakeholders may have access to production records but not transportation details, while others may know the batch number without being able to verify its complete history. Inconsistent information can reduce trust and create disagreements regarding product origin, quantity, or handling.

A common digital platform can connect relevant records and provide controlled access to authorized stakeholders. Blockchain technology can support this process by maintaining a shared, tamper-evident history of important transactions. The system will also include role-based permissions to ensure that users can perform only the activities authorized for their responsibilities.

2.5 Limited Visibility of Honey Quality Information

Quality testing is an important part of honey production and distribution. Relevant information may include laboratory test results, inspection dates, inspector details, and certification status. If these records are maintained separately from the product's batch information, verifying the relationship between a quality report and a specific honey batch can become difficult.

Consumers may also have limited access to quality information before purchasing a product. Producers and retailers may need to spend additional time locating inspection reports or responding to customer questions.

Honey Chain will associate quality inspection records with the corresponding batch identification number. Authorized inspectors will be able to submit test results and verification details. Consumers and other permitted users will be able to view the available quality information through the product's traceability page. This feature will improve the accessibility of documented quality information without treating unverified records as proof of product safety.

3. Problems in Smart Beekeeping Management

In addition to supply-chain traceability, efficient management of beehives is an important requirement for improving honey production operations. Beekeepers need organized information about hive conditions, harvesting activities, and production performance.

3.1 Difficulty in Monitoring Hive Conditions

Beehive conditions can influence colony health and honey production. Depending on the equipment available, useful observations may include temperature, humidity, hive weight, colony activity, and other environmental or operational indicators.

When such information is recorded manually and infrequently, identifying changes in hive conditions may be difficult. Beekeepers may not have a consolidated view of observations from multiple hives, particularly when apiaries are located in different places.

The proposed Honey Chain system will support the recording of hive observations and, where suitable sensors are available, the integration of sensor-generated data. These records can help beekeepers review changes over time and make better-informed management decisions. Sensor-based monitoring is an optional extension and would require compatible hardware, connectivity, and data validation.

3.2 Inaccurate Harvesting and Production Records

Honey harvesting information is essential for understanding production levels and maintaining traceability. The date of harvesting, quantity collected, source hive, and associated batch details must be recorded accurately.

In manual systems, harvesting information may be entered late or associated with the wrong hive. Such mistakes can cause inconsistencies between production records, inventory information, and batch documentation. They can also make it difficult to determine how much honey was collected during a particular period.

Honey Chain will provide a structured harvesting module with mandatory fields and validation rules. Every harvest record will be associated with a registered hive, and each honey batch will be linked to the relevant harvest information. This relationship

will improve record organization and help prevent inconsistencies in the traceability process.

3.3 Limited Production Analysis

Beekeepers may need to compare honey production across hives, harvesting periods, and apiary locations. Without a centralized record system, preparing reports and identifying production patterns can require considerable manual effort.

The proposed system can provide basic reports on harvesting quantities, batch availability, and recorded hive observations. In future versions, analytical features may help users compare production trends and identify unusual changes in recorded data. Such analysis can support planning, although production predictions would require suitable historical data and validation.

4. Need for a Blockchain-Based Solution

Conventional digital databases can organize and retrieve information efficiently, but traceability systems involving multiple independent stakeholders also require mechanisms to preserve record history and establish accountability. If records are modified without an adequate audit trail, users may find it difficult to determine what information was changed and when.

Blockchain technology provides a distributed ledger in which confirmed transactions are linked through cryptographic mechanisms. Depending on the blockchain design and governance model, recorded transactions can be made difficult to alter without detection. This property makes blockchain a potential solution for maintaining important honey supply-chain events.

In Honey Chain, blockchain can be used to record key events such as honey batch creation, ownership or custody transfers, and references to verified inspection records. Each transaction will be associated with a unique batch identifier, timestamp, and authorized participant. Relevant supporting documents may be stored securely outside the blockchain, with suitable references or cryptographic hashes recorded on-chain to support integrity verification.

A permissioned blockchain may be appropriate because beekeepers, producers, inspectors, and distributors have different responsibilities. Access rules can determine who is allowed to submit transactions, verify records, and view specific information.

However, blockchain has certain limitations. It cannot independently verify whether information entered by a user is factually correct. False information entered at the source may still be recorded unless validation and verification procedures are implemented. Blockchain also does not automatically guarantee food safety, honey purity, or regulatory compliance. Therefore, Honey Chain must combine blockchain with user authentication, data validation, inspection procedures, and suitable governance policies.

5. Proposed Honey Chain System

The proposed Honey Chain system will provide an integrated platform for managing beekeeping information and tracking honey through the supply chain. It will connect product records from the original apiary to the final consumer while maintaining appropriate access controls.

5.1 User Registration and Access Control

The system will allow different stakeholders to register or receive authorized accounts according to their roles. Beekeepers will manage hive and harvest records, producers will create honey batches, quality inspectors will enter inspection information, and distributors will record product transfers. Administrators will manage permissions and system operations.

Role-based access control will help protect sensitive information and reduce unauthorized modifications.

5.2 Apiary and Beehive Registration

Beekeepers will be able to create digital records for apiaries and their associated hives. Each hive will have a unique identifier and relevant information, including its location and registration details. Additional hive observations can be recorded when needed.

These records will provide the foundation for connecting harvesting information with the correct source.

5.3 Honey Harvesting and Batch Identification

After harvesting, the beekeeper will enter the harvesting date, quantity, and source hive. The system will validate the submitted information and create a harvest record.

A producer or authorized user will then create a honey batch linked to the corresponding harvest record. The batch will receive a unique identifier that can be used throughout processing, packaging, transportation, and distribution. This approach will reduce ambiguity when several batches are handled simultaneously.

5.4 Blockchain-Based Traceability Records

Important supply-chain events will be recorded in a tamper-evident ledger. These events may include batch registration, processing updates, inspection verification, and transfers between authorized participants.

The system will maintain links between related records so that users can follow a batch's documented history. Previous confirmed events will remain available for audit purposes, subject to the selected blockchain implementation and access policy.

5.5 QR Code-Based Consumer Verification

A QR code will be generated for each registered honey batch or associated retail product. Consumers can scan the code using a smartphone to access the corresponding traceability page.

The page may display the producer, source apiary, harvesting date, processing details, distribution history, and available quality inspection information. The displayed information will depend on what has been recorded and verified. If a code is invalid or the record cannot be found, the system will display an appropriate message.

5.6 Quality and Inventory Management

Authorized quality inspectors will record inspection details and laboratory results against the relevant batch. The system will distinguish between pending, completed, and verified inspection records according to the defined workflow.

Producers and retailers will also be able to maintain inventory information, including batch quantities and stock updates. Connecting inventory records with batch identification will help users organize available stock and maintain more consistent documentation.

6. Expected Outcomes and Significance

The implementation of Honey Chain is expected to provide several benefits to stakeholders across the honey supply chain.

Improved traceability: Linked digital records will make it easier to trace a honey batch back to its documented source and follow its recorded movement through the supply chain.

Greater transparency: Consumers and authorized participants will be able to access relevant product information without relying entirely on disconnected paper records.

Better record management: Digital hive, harvest, batch, and inventory records will reduce dependence on manual documentation and improve information organization.

Improved accountability: Authorized user identities, timestamps, and transaction histories will help establish responsibility for recorded supply-chain activities.

Accessible quality information: Linking inspection reports to batch records will make relevant quality information easier to retrieve and review.

Support for beekeeping decisions: Organized hive observations and harvesting reports will help beekeepers review production history and manage their operations more effectively.

More efficient issue investigation: When a quality concern occurs, batch-level records may help producers identify relevant sources and distribution events more quickly.

These benefits depend on correct data entry, effective verification, user adoption, and reliable system operation. The project aims to improve the management of information and traceability rather than claim that blockchain alone can eliminate adulteration or guarantee product quality.

7. Final Problem Definition

The central problem addressed by this project is the lack of an integrated, transparent, and reliable system that connects beekeeping management with end-to-end honey traceability. Existing manual and disconnected digital processes can make it difficult to maintain accurate hive and harvesting records, verify honey batch origins, share information across supply-chain participants, and provide consumers with accessible product histories.

The proposed Honey Chain system will address these challenges through blockchain-based traceability, digital beekeeping records, unique honey batch identification, QR code verification, quality inspection management, and inventory tracking. It will establish links between the source hive, harvesting event, honey batch, quality records, and distribution activities. Appropriate access controls and verification procedures will support data integrity and accountability.

The ultimate goal is to develop a practical software platform that improves the transparency and organization of honey production and distribution while supporting beekeepers in managing their activities. By connecting reliable digital records with tamper-evident blockchain transactions, Honey Chain aims to strengthen trust among producers, distributors, retailers, and consumers and contribute to a more traceable and efficiently managed honey supply chain.